Signer Window Cropping with FFmpeg

This project demonstrates how to crop the **sign language interpreter window** from Sri Lankan Parliament broadcast videos using **FFmpeg**.
The cropped videos are intended for building a **sign language recognition dataset**.

1. Requirements

- FFmpeg installed on your system
- A video file (e.g., `video.mp4`)
- Terminal or VS Code terminal

Download FFmpeg:
https://ffmpeg.org/download.html

2. Check Video Information

Run the following command to view the video resolution and metadata.

```
ffmpeg -i video.mp4
```

Example output:

```
1920x1080, 30 fps
```

3. Cropping Method

The **crop filter** in FFmpeg was used to extract the interpreter region.
The crop filter uses the following format:

```
crop=width:height:x:y
```

Where:

Parameter	Description
width	Width of the cropped region
height	Height of the cropped region
x	Horizontal starting position from the left
y	Vertical starting position from the top

This allows selecting a specific rectangular region of the video.

4. Preview Cropping

Before applying the crop to the entire video, a **5-second preview** was generated to verify the selected region.

The following command was used:

```
ffmpeg -t 5 -i video.mp4 -vf "crop=240:205:1580:700" preview.mp4
```

Explanation

Command Part	Function
`-t 5`	Extract only the first 5 seconds for preview
`-i video.mp4`	Input video file
`-vf`	Apply video filter
`crop=240:205:1580:700`	Cropping parameters
`preview.mp4`	Output preview video

This step ensures that the correct interpreter region is selected before processing the full video.

5. Final Cropping Command

After confirming the correct crop region, the crop operation was applied to the entire video.

```
ffmpeg -i video.mp4 -vf "crop=240:205:1580:700" signer.mp4
```

Parameter Meaning

Parameter	Value	Description
width	240 pixels	Width of interpreter window
height	205 pixels	Height of interpreter window
x	1580 pixels	Horizontal starting point
y	700 pixels	Vertical starting point

This command extracts only the interpreter region and saves it as a new video file.

6. Result

The resulting video contains only the **sign language interpreter**, removing unnecessary areas such as:

- Main speaker
- Background elements
- Bottom information panels

The cropped video retains important visual features required for **sign language recognition**, including:

- Face
- Upper body
- Both hands
- Signing gestures

7. Importance for Sign Language Processing

Cropping the interpreter region improves the efficiency of further processing tasks such as:

- Gesture detection
- Hand tracking
- Sign language recognition
- Machine learning model training

By removing irrelevant parts of the frame, the computational cost is reduced and model accuracy can improve.

8. Summary

The FFmpeg crop filter was successfully used to extract the sign language interpreter region from the full broadcast video. A preview step was first performed to determine the correct crop coordinates, followed by applying the final cropping command to the entire video.

The selected crop parameters:

```
crop=240:205:1580:700
```

accurately isolate the interpreter window while preserving essential signing information.